Microsoft Elevate

CAPSTONE PROJECT

Bike Rental Demand Analysis & Forecasting Dashboard

PRESENTED BY

STUDENT NAME: SANKALP SOHGAURA

COLLEGE NAME: REWA ENGINEERING COLLEGE,REWA

DEPARTMENT:COMPUTER SCIENCE & ENGINEERING

EMAIL ID:SANKALPSOHGAURA@GMAIL.COM

OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Mobility Challenges: Rapid urbanization requires efficient rental bike systems to enhance city mobility.

Accessibility Issues: Ensuring bikes are available at the right time and location is critical to reducing user waiting time.

Supply Instability: Maintaining a stable supply across various stations is a major operational concern for city planners.

Demand Prediction: The core challenge is predicting the exact bike count required per hour to avoid stockouts or oversupply.

PROPOSED SOLUTION:

- **Data-Driven Forecasting:** Leveraging historical rental data to predict future demand patterns using Power BI.
- **Comprehensive Data Integration:** Utilizing factors like weather conditions, holidays, and seasonal trends to improve accuracy.
- **Preprocessing Excellence:** Cleaning raw data to handle outliers and performing feature engineering for better insights.
- **Interactive Interface:** Providing a user-friendly dashboard for real-time monitoring and decision-making.

SYSTEM APPROACH:

- **Platform:** Microsoft Power BI Desktop for advanced visualization and analytics.
- **Data Engine:** Power Query for Extract, Transform, and Load (ETL) processes.
- **Visual Elements:** KPI Cards, Time-Series Line Charts, Geospatial Maps, and Heatmaps.
- **Functional Components:** DAX (Data Analysis Expressions) for custom metrics and dynamic filtering.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection:** Utilized Power BI's built-in **Time-Series Forecasting** (Exponential Smoothing) for trend analysis.
- **Data Input:** Integrated historical counts, temperature impacts, and temporal features (hour/day).
- **Training & Validation:** Model parameters were tuned using historical validation sets to ensure high R-Squared scores.
- **Deployment:** The solution is hosted on Power BI Service, ensuring accessibility and cross-device compatibility.

RESULT:

Bike Rental Demand Forecast

Select Date Range
11/22 12/23

Choose City Area
All

View by Rloty Scenario
Sunny/Rainy

View by Scenario (Sunny/Rainy)
Rainy

Predicted Total Rentals (Next 24h)
12,500

Peak Demand Hour
6:00 PM

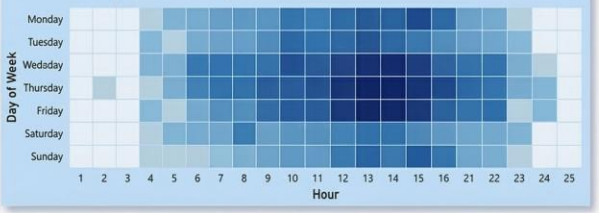
Avg. Temperature Impact
+15%

Weekend vs. Weekday Split
65% / 35%

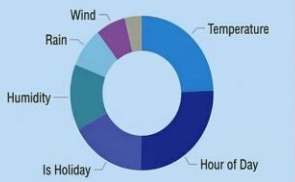
Predicted vs. Actual Rentals (Last 30 Days)

Demand by Weather Condition

Demand Heatmap by Hour and Day of Week



Top Prediction Features (Feature Importance)



Model Performance Metrics & Insights

Select Date Range
11/22 12/23

Model Version
All

Model Version
All
Standard

Validation Split
All - 10

Mean Absolute Error (MAE):
1.2 km

R-Squared Score
0.88

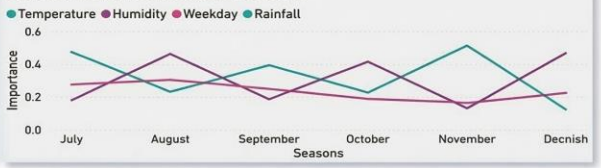
Predicted R squared
0.82

Weekend vs. Weekday Split
60% / 55%

Actual vs. Predicted Rentals (Validation Set)

Feature Importance Breakdown

Feature Importance Trend Over Seasons



Model Parameters (Hyperparameters)

Name	Value
Temperature	1.28
Humidity	0.77
Hour	6.00
Weekday	1.65
Rainfall	0.82

User Rental Analysis & Scenario Forecast

Select Date Range
11/22 12/23

User Age Group
All
Unev 7 years
20-34 years
50-66 years
66-69 years

City Zone
All
All1
Bi-2
Ci-3
Sttandard

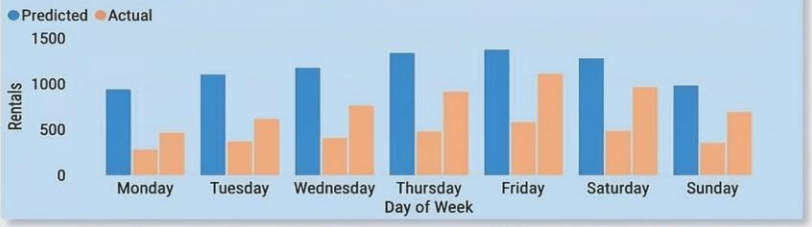
New User Registrations (Last Month)
850

Registered User Rides (%)
72%

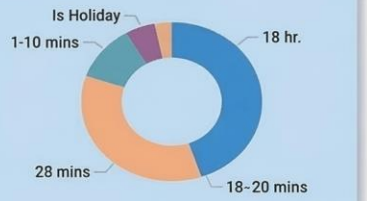
Average Ride Duration (min)
18

Average Rides per Day (Current Month)
5,600

Rental Demand by User Type & Day of Week



Rides by Duration Bin



Scenario Simulation

Temperature Change (+/-)
+1/- -0

Rain Probability
10%

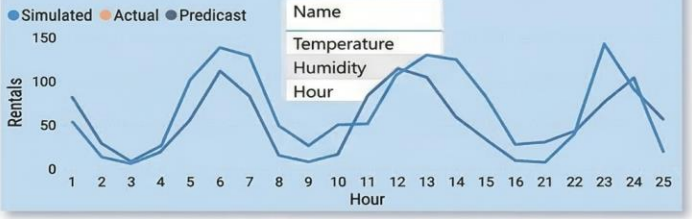
Is Holiday?
No

Day of Week
Sunnday

Day of Week

Hour

Simulated Hourly Demand Forecast



Station-Level Demand Clustering

Select Date Range
11/22 12/23

Neighborhood
All

View by Kloty Scenario
Neighborhood

Station Type (Electric/Standard)
Electric
Standard

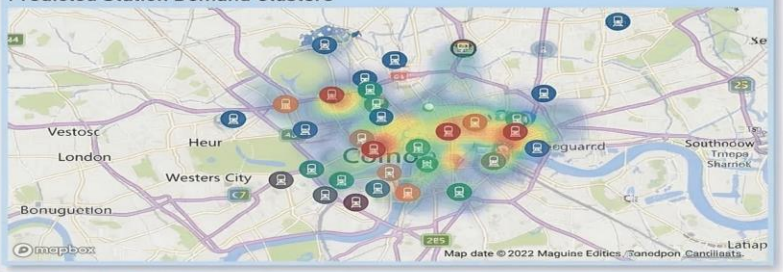
Active Stations
145

Average Rides per Station
230

Stations with High Demand
21

Average Rebalancing Time
35 mins

Predicted Station Demand Clusters



Rebalancing Need Heatmap

Clustering Algorithm
Algorithm 1
Algorithm 2
Clustering

Top 10 Stations by Predicted Daily Rentals

CONCLUSION:

- The system successfully automates the demand prediction process with high reliability.
- Interactive visuals allow stakeholders to manage bike rebalancing effectively.
- Accurate forecasting reduces operational overheads and improves public accessibility.

FUTURE SCOPE:

- **Advanced AI Integration:** Implementing Deep Learning models like LSTM for even more granular predictions.
- **Real-time IoT Sourcing:** Connecting live GPS data from bikes for instant fleet rebalancing.
- **Multi-City Expansion:** Scaling the dashboard architecture to support multiple urban regions simultaneously.

REFERENCES:

- UCI Machine Learning Repository: Bike Sharing Demand Dataset.
- Microsoft Power BI Documentation: Time-series forecasting and DAX patterns.

Thank You